

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: Task 1(part A)

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Date: 30-08-2026

1. TITLE

Design and Develop a College Information Portal Webpage Using HTML and HTML5

2. AIM

To design and develop a static webpage titled “College Information Portal” using HTML and HTML5 that demonstrates the fundamental building blocks of web page creation, including document structure, text formatting, hyperlinks, lists, tables, images, forms, semantic elements, HTML5 input types, and audio/video embedding.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and apply the basic structure of an HTML5 document.
- To display information using headings and paragraphs with appropriate text formatting (Bold, Italic, Underline, Superscript, Subscript).
- To create ordered and unordered lists.
- To insert hyperlinks to internal sections and external websites.
- To design a table for displaying structured student data.
- To insert and describe images using the alt attribute.
- To design a registration form using various input controls (text, email, password, radio, checkbox, dropdown, text area).
- To use HTML5 semantic elements (header, nav, section, article, footer) for meaningful page structure.
- To apply HTML5 input types such as Email, Date, Number, Color and Range.
- To embed audio and video content using HTML5 media elements.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages. An HTML5 document begins with the `<!DOCTYPE html>` declaration and is organized using nested elements (tags) that define headings, paragraphs, lists, tables, images, links, forms and media.

HTML5 introduced semantic elements such as `<header>`, `<nav>`, `<section>`, `<article>` and `<footer>`, which describe the meaning of the content they contain rather than just its appearance. This improves readability, accessibility and search engine optimization compared to using generic `<div>` elements everywhere.

Forms are created using the `<form>` element and various `<input>` types. HTML5 added new input types — email, date, number, color and range — which provide built-in validation and richer controls in modern browsers without needing extra JavaScript.

HTML5 also introduced native `<audio>` and `<video>` elements, allowing media to be embedded directly in a page with built-in playback controls, removing the earlier dependency on third-party plugins such as Flash.

5. ALGORITHM / PROCEDURE

1. Create a new file named index.html.
2. Declare the document type using <!DOCTYPE html> and set up the <html>, <head> and <body> structure.
3. Build a semantic <header> containing the college logo and college name.
4. Add a <nav> bar with links to each section of the page and to external sites.
5. Write an “About” <section> using headings, paragraphs and text-formatting tags (bold, italic, underline, superscript, subscript).
6. Create one unordered list (departments offered) and one ordered list (admission steps).
7. Build a <table> to display student Roll No, Name, Branch and Email.
8. Insert the college logo and a campus image, each with a descriptive alt attribute.
9. Design a registration <form> with text, email, password, date, number, radio, checkbox, dropdown, color, range and textarea inputs, plus submit and reset buttons.
10. Embed an audio file and a video file using the <audio> and <video> elements.
11. Add a semantic <footer> with copyright and contact information.
12. Save the file and open it in a web browser.
13. Verify that every heading, list, link, table, image, form control and media element renders and behaves as expected.
14. Record the observations against the expected results.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal - ANITS</title>
</head>
<body>

  <!-- PART D: HTML5 Semantic Header -->
  <header>
    <table width="100%">
      <tr>
        <td width="120">
          
        </td>
        <td>
          <h1 align="center">ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES</h1>
          <h3 align="center">(AUTONOMOUS)</h3>
          <p align="center"><em>Autonomous Institution | Accredited by NBA & NAAC with 'A+'
Grade</em></p>
          <p align="center">Sangivalasa, Bheemunipatnam, Visakhapatnam, Andhra Pradesh</p>
        </td>
      </tr>
    </table>
  </header>

  <!-- PART D: Navigation Bar -->
  <nav align="center">
    <a href="#about">About ANITS</a> |
    <a href="#academics">Academics</a> |
    <a href="#students">Student Data</a> |
    <a href="#register">Admission Registration</a> |
    <a href="#location">Location</a> |
    <a href="https://anits.edu.in" target="_blank">ANITS Official Website</a> |
    <a href="https://www.google.com" target="_blank">Google Search</a>
```

```

</nav>

<hr>

<main>
  <!-- PART A: Basic HTML & Text Formatting -->
  <section id="about">
    <h2>About ANITS</h2>
    <article>
      <p>
        <strong>Anil Neerukonda Institute of Technology and Sciences
        (ANITS)</strong> was established in the academic year <i>2001-02</i> by the Anil
        Neerukonda Educational Society (ANES).
        The college is permanently affiliated with <u>Andhra University</u>
        and approved by AICTE.
      </p>
      <p>
        <strong>Text formatting demonstration:</strong>
        ANITS celebrated its 25<sup>th</sup> anniversary of academic
        excellence in 2026, and
        its research labs regularly conduct experiments involving compounds
        such as H<sub>2</sub>O and CO<sub>2</sub>.
      </p>
    </article>
  </section>

  <!-- PART A: Lists & Hyperlinks -->
  <section id="academics">
    <h2>Academic Overview</h2>

    <h3>Offered Departments (Unordered List)</h3>
    <ul>
      <li>Computer Science & Engineering (CSE)</li>
      <li>CSE (Artificial Intelligence & Machine Learning)</li>
      <li>Electronics & Communication Engineering (ECE)</li>
      <li>Electrical & Electronics Engineering (EEE)</li>
      <li>Information Technology (IT)</li>
      <li>Mechanical Engineering</li>
    </ul>

    <h3>EAPCET / APICET Admission Steps (Ordered List)</h3>
    <ol>
      <li>Qualify in the AP EAPCET / ICET entrance examination.</li>
      <li>Register for web counseling and select <strong>ANIL</strong> as
        college code.</li>
      <li>Certificate verification at designated helpline centers.</li>
      <li>Report to ANITS campus upon seat allotment and complete fee
        payment.</li>
    </ol>

    <p>
      To learn more about campus achievements, visit:
      <a href="https://anits.edu.in" target="_blank">ANITS Official Portal</a>
    </p>
  </section>

  <hr>

  <!-- PART B: Tables and Images -->
  <section id="students">
    <h2>ANITS Top Rankers (Student Details)</h2>

    <table border="1" cellpadding="8" cellspacing="0">

```

```

<thead>
  <tr>
    <th>Roll No</th>
    <th>Student Name</th>
    <th>Branch</th>
    <th>Email ID</th>
  </tr>
</thead>
<tbody>
  <tr>
    <td>321126510001</td>
    <td>K. Sitaram</td>
    <td>CSE</td>
    <td>sitaram.cse@anits.edu.in</td>
  </tr>
  <tr>
    <td>321126512015</td>
    <td>P. Sravani</td>
    <td>ECE</td>
    <td>sravani.ece@anits.edu.in</td>
  </tr>
  <tr>
    <td>321126520042</td>
    <td>V. Rahul</td>
    <td>IT</td>
    <td>rahul.it@anits.edu.in</td>
  </tr>
</tbody>
</table>

<h3>Campus Overview</h3>

</section>

<hr>

<!-- PART C & D: Forms & Modern HTML5 Input Types -->
<section id="register">
  <h2>ANITS Admission Registration Form</h2>
  <form action="#" method="post">

    <p>
      <label for="fullname">Student Full Name:</label><br>
      <input type="text" id="fullname" name="fullname" placeholder="Enter
full name" required>
    </p>

    <p>
      <label for="email">Student Email ID:</label><br>
      <input type="email" id="email" name="email"
placeholder="student@anits.edu.in" required>
    </p>

    <p>
      <label for="password">Create Password:</label><br>
      <input type="password" id="password" name="password" required>
    </p>

    <p>
      <label for="dob">Date of Birth:</label><br>
      <input type="date" id="dob" name="dob">
    </p>
  </form>
</section>

```

```

    </p>

    <p>
      <label for="eapcet_rank">AP EAPCET Rank:</label><br>
      <input type="number" id="eapcet_rank" name="eapcet_rank" min="1"
max="200000">
    </p>

    <p>
      <label>Gender:</label><br>
      <input type="radio" id="male" name="gender" value="Male">
      <label for="male">Male</label>
      <input type="radio" id="female" name="gender" value="Female">
      <label for="female">Female</label>
    </p>

    <p>
      <label>Co-Curricular / Hobbies:</label><br>
      <input type="checkbox" id="coding" name="hobbies" value="Coding">
      <label for="coding">Coding Club</label>
      <input type="checkbox" id="robotics" name="hobbies" value="Robotics">
      <label for="robotics">e-Yantra Robotics</label>
      <input type="checkbox" id="sports" name="hobbies" value="Sports">
      <label for="sports">Sports & Athletics</label>
    </p>

    <p>
      <label for="branch">Select Engineering Branch:</label><br>
      <select id="branch" name="branch">
        <option value="CSE">Computer Science & Engineering (CSE)</option>
        <option value="CSM">CSE (AI & ML)</option>
        <option value="ECE">Electronics & Communication (ECE)</option>
        <option value="EEE">Electrical & Electronics (EEE)</option>
        <option value="IT">Information Technology (IT)</option>
        <option value="MECH">Mechanical Engineering</option>
      </select>
    </p>

    <p>
      <label for="fav_color">Preferred ID Card Theme Color:</label><br>
      <input type="color" id="fav_color" name="fav_color" value="#003366">
    </p>

    <p>
      <label for="satisfaction">Self Assessment Rating (1 to
10):</label><br>
      <input type="range" id="satisfaction" name="satisfaction" min="1"
max="10">
    </p>

    <p>
      <label for="comments">Additional Remarks / Address:</label><br>
      <textarea id="comments" name="comments" rows="4" cols="50"
placeholder="Enter address or questions..."></textarea>
    </p>

    <p>
      <input type="submit" value="Submit Registration">
      <input type="reset" value="Reset Form">
    </p>
  </form>
</section>

```

```

<hr>

<!-- Additional: Iframe (Frames/Iframes concept) -->
<section id="location">
  <h2>Campus Location</h2>
  <iframe
    title="ANITS Location Map"
    src="https://www.google.com/maps?
q=ANITS+Sangivalasa+Bheemunipatnam&output=embed"
    width="600" height="300" style="border:0;">
  </iframe>
</section>

<hr>

<!-- PART D: Audio & Video Elements -->
<section id="media">
  <h2>Campus Media</h2>

  <h3>ANITS College Anthem (Audio)</h3>
  <audio controls>
    <source src="college-anthem.mp3" type="audio/mpeg">
    Your browser does not support the audio element.
  </audio>

  <br><br>

  <h3>ANITS Campus Tour (Video)</h3>
  <video width="600" controls>
    <source src="campus-tour.mp4" type="video/mp4">
    Your browser does not support the video tag.
  </video>
</section>

</main>

<hr>

<!-- PART D: Semantic Footer -->
<footer>
  <p>&copy; 2026 Anil Neerukonda Institute of Technology & Sciences (ANITS). All
Rights Reserved.</p>
  <p>Contact: principal@anits.edu.in | Phone: 08933-225083</p>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Tag / Attribute	Description
<!DOCTYPE html>	Declares the document as an HTML5 document.
<header>, <nav>, <section>, <article>, <footer>	HTML5 semantic elements that structure the page into meaningful regions.
, , <u>	Apply bold, italic and underline text formatting respectively.
<sup>, <sub>	Render superscript and subscript text (e.g. 25th, H2O).
, ,	Create unordered and ordered lists of items.
	Creates a hyperlink to an internal section or an external website.
<table>, <thead>, <tbody>, <tr>, <th>, <td>	Build a structured table of student details.
	Embeds an image and provides alternate text for accessibility.
<iframe>	Embeds external content (a map) inside the page.
<form>, <input>, <select>, <textarea>	Collect user input through various form controls.
type="radio" / "checkbox"	Allow single-choice (gender) and multiple-choice (hobbies) selection.
type="email", "date", "number", "color", "range"	HTML5 input types that provide built-in validation and richer controls.
<audio controls>, <video controls>	Embed playable audio and video content with native browser controls.

8. TEST CASES AND EXECUTION DATA

The page was opened in a browser and each feature was individually verified:

Test Case	Feature Tested	Action Performed	Expected Result	Status
TC-01	Page structure	Open index.html in browser	Header, nav, all sections and footer render correctly	Pass
TC-02	Text formatting	View About section	Bold, italic, underline, superscript and subscript text display correctly	Pass
TC-03	Lists	View Academics section	Unordered department list and ordered admission-steps list display correctly	Pass
TC-04	Hyperlinks	Click ANITS Official Website and Google Search links	Each link opens the correct site in a new tab	Pass
TC-05	Table	View Student Details table	Roll No, Name, Branch and Email display in a structured table	Pass
TC-06	Images	View logo and campus image	Both images render with correct alt text on hover / load failure	Pass
TC-07	Form validation	Submit form with Name/Email	Browser blocks submission and	Pass

Test Case	Feature Tested	Action Performed	Expected Result	Status
		left empty	highlights required fields	
TC-08	Radio buttons	Select Male / Female	Only one gender option can be selected at a time	Pass
TC-09	Checkboxes	Select multiple hobbies	Multiple hobby options can be selected simultaneously	Pass
TC-10	Dropdown	Select a branch from the list	Selected branch value updates correctly	Pass
TC-11	Color & Range inputs	Pick a theme color and drag the rating slider	Color swatch and slider value update live	Pass
TC-12	Date & Number inputs	Pick a date of birth and enter EAPCET rank	Date picker opens; number field rejects non-numeric input	Pass
TC-13	Audio / Video	Play embedded audio and video	Native browser controls appear and media plays (sample media files)	Pass
TC-14	Iframe	View Campus Location section	Embedded map loads inside the page	Pass

9. OUTPUT



ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES

(AUTONOMOUS)

Autonomous Institution | Accredited by NBA & NAAC with 'A+' Grade

Sangivalasa, Bheemunipatnam, Visakhapatnam, Andhra Pradesh

[About ANITS](#) | [Academics](#) | [Student Data](#) | [Admission Registration](#) | [ANITS Official Website](#) | [Google Search](#)

About ANITS

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1. Qualify in the AP EAPCET / ICET entrance examination.
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ANITS Top Rankers (Student Details)

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321126512015	P. Sravani	ECE	sravani.ece@anits.edu.in
321126520042	V. Rahul	IT	rahul.it@anits.edu.in

Campus Overview

 ANITS Sangivalasa Campus Building

ANITS Admission Registration Form

Student Full Name:

Student Email ID:

Create Password:

Date of Birth:

AP EAPCET Rank:

Gender:
☐ Male ☐ Female

Co-Curricular / Hobbies:
☐ Coding Club ☐ e-Yantra Robotics ☐ Sports & Athletics

Select Engineering Branch:

Preferred ID Card Theme Color:

Self Assessment Rating (1 to 10):

Additional Remarks / Address:

Campus Media

ANITS College Anthem (Audio)

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0:00 / 0:00

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ANITS Campus Tour (Video)

▶

0:00

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10. RESULT

Thus, a webpage titled “College Information Portal” was successfully designed and developed using HTML and HTML5. Basic HTML tags, text formatting, ordered and unordered lists, hyperlinks, tables, images, a registration form with multiple input types, HTML5 semantic elements, and audio/video embedding were implemented and verified through browser testing.

